

3. Salam exercises after every **10** days and Nadeem every **6** days. Salam and Nadeem both exercised today. After how many days will they be unite and exercise together again?
4. Hooria is thinking of a number that is divisible by both **15** and **21**. What is the smallest possible number that Hooria could be thinking of?
5. Four vans leave Sukkur at the same time. The first van is seen after 3 hours, the second after 4 hours, the third after 5 hours and the fourth after 6 hours. After how many hour from the starting time, will all of them come together.
6. Boxes that are **12** cm tall are being stacked next to boxes that are **18** cm tall. What is the shortest height at which the two stacks will be of the same height?
7. Samina has two peices of cloth. One piece is **64** cm wide and the other piece is **80** cm wide. She wants to cut both pieces into strips of equal width that are as wide as possible. How wide should she cut the strips?
8. Three milk container hold 30 litres, 40 litres and 50 litres respectively. Find the capacity of a measuring utensils which can hold the milk in each container an exact number of times.
9. Find the greatest length of a measure of stick which can be used to measure exactly **30 cm**, **60 cm** or **90 cm**.
10. Pencils are in packages of **10**. Erasers are in packages of **12**. A shopkeeper wants to purchase the smallest number of pencils and erasers so that he will have exactly **1** eraser per pencil. How many packages of pencils and erasers should shopkeeper buy?

REVIEW EXERCISE 2

1. Tick (✓) the correct answer.

(i) HCF of 2, 4, 10 is:

- (a) 2 (b) 4 (c) 10 (d) 1

(ii) LCM of 3, 6, 9 is:

- (a) 3 (b) 6 (c) 9 (d) 18

(iii) HCF of 3, 5, 7 is:

- (a) 3 (b) 1 (c) 7 (d) 5

(iv) LCM of 2, 3 and 5 is:

- (a) 6 (b) 10 (c) 30 (d) 15

(v) To find HCF by division method we divide greater number by _____.

- (a) same number (b) twice the number
(c) smaller number (d) greater number

2. Find the least number which is exactly divisible by 6, 8 and 12.
3. Find the greatest number which divides 12 and 18 exactly.
4. What is the least number of apples which can be distribute equally among 10, 15 or 20 children.
5. Two pieces of cloth 15 metres and 20 metres long respectively are to be cut into small pieces of equal lengths. What will be the greatest length of each piece?
6. What will be capacity of smallest drum which can be filled completely using each of the utensils that can fill exactly 15/, 25/, 40/ or 75/.

3.1 ADDITION AND SUBTRACTION

3.1.1 Add and subtract two and more fractions with different denominators.

(a) Addition of fractions

We have learnt in previous class about the addition and subtraction of two fractions. When we add two fractions with different denominators, first we will make the denominators equal to each other by finding the equivalent fractions.

Let us see the following examples:

Example 1. Add $\frac{2}{5}$ and $\frac{1}{4}$

Method 1. By making equivalent fractions

Solution:

First we find the equivalent fractions of $\frac{2}{5}$ and $\frac{1}{4}$

Finding equal multiples of 5 and 4.

$$\text{So, } \frac{2}{5} = \frac{2 \times 4}{5 \times 4} = \frac{8}{20} \text{ and } \frac{1}{4} = \frac{1 \times 5}{4 \times 5} = \frac{5}{20}$$

Now, we have a common denominator of both the fractions.

$$\text{So, } \frac{2}{5} + \frac{1}{4} = \frac{8}{20} + \frac{5}{20} = \frac{8+5}{20} = \frac{13}{20}$$

Sum of fractions = Sum of their equivalent fractions

Teacher's Note

Teacher should revise addition and subtraction of two fractions with different denominators and learn them with more fractions also.

Method 2. By finding LCM of denominators

$$\frac{2}{5} + \frac{1}{4}$$

Find LCM of the denominators

$$\begin{aligned}\frac{2}{5} + \frac{1}{4} &= \frac{(2 \times 4) + (1 \times 5)}{20} \\ &= \frac{8 + 5}{20} \\ &= \frac{13}{20}\end{aligned}$$

2	5, 4
2	5, 2
5	5, 1
	1, 1

$$\text{LCM} = 2 \times 2 \times 5 = 20$$

$$20 \div 5 = 4, 20 \div 4 = 5$$

So, dividing LCM 20 by denominator of $\frac{2}{5}$, $20 \div 5 = 4$
Dividing 20 by denominator of $\frac{1}{4}$, $20 \div 4 = 5$

Example 1. Solve: $\frac{1}{4} + \frac{1}{6} + \frac{3}{8}$

Solution: $\frac{1}{4} + \frac{1}{6} + \frac{3}{8}$

First find LCM of the denominators.

The LCM of 4, 6, 8 is 24

$$\text{LCM} = 2 \times 2 \times 2 \times 3 = 24$$

$$\frac{1}{4} + \frac{1}{6} + \frac{3}{8} = \frac{(1 \times 6) + (1 \times 4) + (3 \times 3)}{24}$$

$$= \frac{6 + 4 + 9}{24}$$

$$= \frac{19}{24}$$

Hence, $\frac{1}{4} + \frac{1}{6} + \frac{3}{8} = \frac{19}{24}$

2	4, 6, 8
2	2, 3, 4
2	1, 3, 2
3	1, 3, 1
	1, 1, 1

$$\begin{aligned}24 \div 4 &= 6 \\ 24 \div 6 &= 4 \\ 24 \div 8 &= 3\end{aligned}$$

So, dividing LCM 24 by denominator of $\frac{1}{4}$, $24 \div 4 = 6$
Dividing 24 by denominator of $\frac{1}{6}$, $24 \div 6 = 4$
Dividing 24 by denominator of $\frac{3}{8}$, $24 \div 8 = 3$

(b) Subtraction of two fractions.

Example 1: Subtract $\frac{2}{7}$ from $\frac{1}{3}$ **Solution:** To subtract $\frac{2}{7}$ from $\frac{1}{3}$ means to solve $\frac{1}{3} - \frac{2}{7}$ **Method 1.** By making equivalent fractions:First we find the equivalent fractions of $\frac{1}{3}$ and $\frac{2}{7}$

$$\frac{1}{3} = \frac{1}{3} \times \frac{7}{7} = \frac{1 \times 7}{3 \times 7} = \frac{7}{21}$$

$$\text{and } \frac{2}{7} = \frac{2}{7} \times \frac{3}{3} = \frac{2 \times 3}{7 \times 3} = \frac{6}{21}$$

Now we have a common denominator of both the fractions.

$$\begin{aligned} \text{So, } \frac{1}{3} - \frac{2}{7} &= \frac{7}{21} - \frac{6}{21} \\ &= \frac{7-6}{21} = \frac{1}{21} \end{aligned}$$

Difference of two fractions = Difference of their equivalent fractions.

Method 2. Finding LCM of denominators:**Solution:** First find LCM of the denominators 3 and 7.

Now,

$$\frac{1}{3} - \frac{2}{7} = \frac{(1 \times 7) - (2 \times 3)}{21}$$

$$= \frac{7-6}{21}$$

$$= \frac{1}{21}$$

$$\text{Thus, } \frac{1}{3} - \frac{2}{7} = \frac{1}{21}$$

3	3, 7
7	1, 7
	1, 1

$$\text{LCM} = 3 \times 7 = 21$$

So, divide 21 by denominator of $\frac{1}{3}$, $21 \div 3 = 7$ Divide 21 by denominator of $\frac{2}{7}$, $21 \div 7 = 3$

Example 2. Solve: $\frac{14}{5} - 2\frac{3}{4}$

Solution:

First we change the mixed fraction into an improper fraction.

$$\begin{aligned}\text{Now, } & \frac{14}{5} - 2\frac{3}{4} \\ &= \frac{14}{5} - \frac{11}{4}\end{aligned}$$

Then find the LCM

$$\begin{aligned}\frac{14}{5} - 2\frac{3}{4} &= \frac{(14 \times 4) - (11 \times 5)}{20} \\ &= \frac{56 - 55}{20} = \frac{1}{20}\end{aligned}$$

$$\text{Thus, } \frac{14}{5} - 2\frac{3}{4} = \frac{1}{20}$$

2	5, 4
2	5, 2
5	5, 1
	1, 1

$$\text{LCM} = 2 \times 2 \times 5 = 20$$

Example 3. Solve: $\frac{4}{3} + \frac{1}{2} - \frac{1}{8}$

Solution:

$$\frac{4}{3} + \frac{1}{2} - \frac{1}{8}$$

First we find the LCM of 2, 3 and 8

$$\text{LCM} = 2 \times 2 \times 2 \times 3 = 24$$

Therefore,

$$\begin{aligned}&= \frac{(4 \times 8) + (1 \times 12) - (1 \times 3)}{24} \\ &= \frac{32 + 12 - 3}{24} = \frac{41 - 3}{24} \\ &= \frac{38}{24} = 2\frac{17}{24}\end{aligned}$$

$$\text{Thus, } \frac{4}{3} + \frac{1}{2} - \frac{1}{8} = 2\frac{17}{24}$$

2	3, 2, 8
2	3, 1, 4
2	3, 1, 2
3	3, 1, 1
	1, 1, 1

$$\text{LCM} = 2 \times 2 \times 2 \times 3 = 24$$

EXERCISE 3.1

A. Add the following:

(1) $\frac{1}{3} + \frac{1}{2}$ (2) $\frac{3}{4} + \frac{1}{8}$ (3) $\frac{2}{5} + \frac{1}{3}$
(4) $\frac{3}{8} + \frac{1}{3}$ (5) $\frac{2}{9} + \frac{3}{4}$ (6) $\frac{2}{5} + \frac{3}{7}$

B. Solve:

(1) $\frac{1}{3} + \frac{1}{5} + \frac{1}{9}$ (2) $\frac{1}{6} + \frac{1}{15} + \frac{1}{18}$ (3) $\frac{1}{8} + \frac{1}{12} + \frac{1}{16}$
(4) $1\frac{1}{10} + 1\frac{1}{5} + \frac{2}{20}$ (5) $2\frac{1}{3} + 1\frac{1}{15} + 1\frac{1}{20}$ (6) $1\frac{1}{24} + \frac{1}{32} + 1\frac{1}{4}$

C. Subtract the following:

(1) $\frac{3}{4} - \frac{3}{8}$ (2) $\frac{2}{3} - \frac{1}{4}$ (3) $\frac{3}{4} - \frac{1}{2}$
(4) $\frac{5}{6} - \frac{1}{2}$ (5) $4\frac{2}{5} - 1\frac{1}{4}$ (6) $3\frac{3}{5} - 2\frac{9}{10}$

D. Solve:

(1) $\frac{7}{8} - \frac{1}{3} - \frac{1}{4}$ (2) $\frac{3}{4} - \frac{1}{6} - \frac{1}{3}$ (3) $\frac{6}{7} - \frac{1}{14} - \frac{1}{2}$
(4) $2\frac{5}{6} - 2\frac{1}{3} - \frac{1}{4}$ (5) $2\frac{11}{12} - 2\frac{1}{6} - 1\frac{1}{4}$ (6) $2\frac{9}{10} - 3\frac{1}{2} - 2\frac{1}{5}$
(7) $4\frac{8}{9} - 3\frac{1}{6} - 4\frac{1}{3}$ (8) $3\frac{1}{12} - 3\frac{1}{4} - 2\frac{1}{6}$ (9) $2\frac{3}{15} - 1\frac{2}{5} - 1\frac{3}{10}$

E. Add:

(1) $\frac{2}{5}$ and $\frac{3}{10}$ (2) $\frac{1}{3}$ and $\frac{1}{6}$ (3) $\frac{1}{7}$, $\frac{1}{14}$ and $\frac{1}{21}$

F. Subtract:

(1) $\frac{2}{3}$ from $\frac{3}{4}$ (2) $\frac{1}{3}$ from $\frac{4}{5}$ (3) $\frac{3}{4}$ from $\frac{4}{5}$

3.2 MULTIPLICATION

3.2.1 Multiply a fraction by a number and demonstrate with the help of diagram

Let us see the following example:

Example 1. Multiply $\frac{2}{3}$ by 3 using diagram.

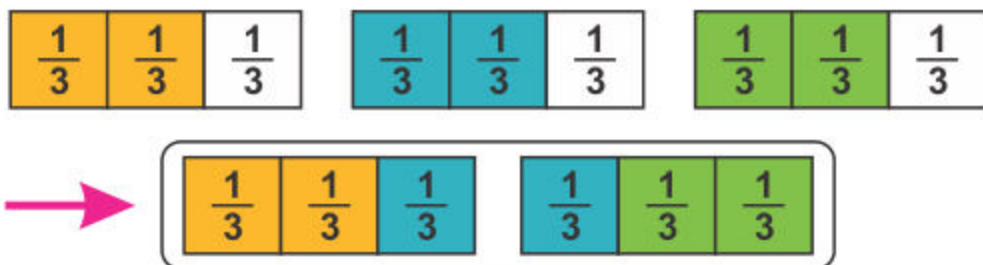
Solution:

As we know that multiply $\frac{2}{3}$ by 3 is expressed in symbol as $\frac{2}{3} \times 3$

We can express the multiplication as a repeated addition.

$$\frac{2}{3} \times 3 = 3 \times \frac{2}{3} = \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = \frac{2+2+2}{3} = \frac{6}{3}$$

With the help of diagram, we can show as follows:



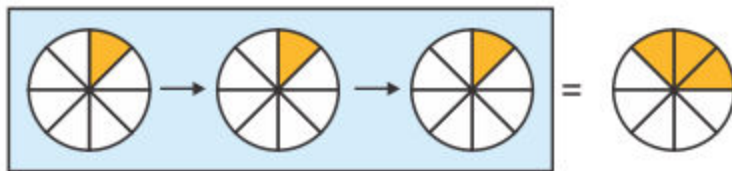
The coloured parts are $\frac{6}{3}$ or 2 whole.



Activity

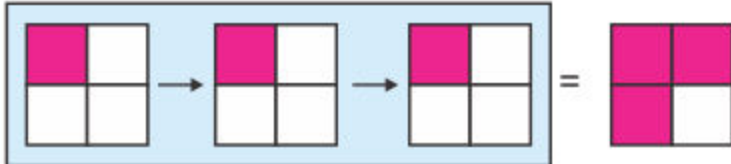
Use the diagrams to solve the following:

(1) $\frac{1}{8} \times 3$



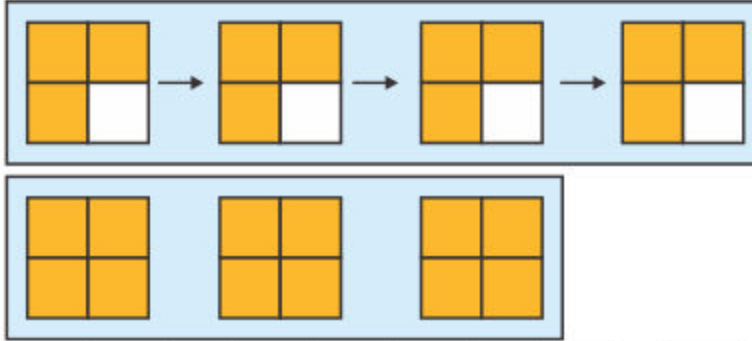
$$\text{So, } \frac{1}{8} \times 3 = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{8}$$

(2) $\frac{1}{4} \times 3$



So, $\frac{1}{4} \times 3 = \square + \square + \square = \square$

(3) $\frac{3}{4} \times 4$



So, $\frac{3}{4} \times 4 = \square + \square + \square + \square = \square$

To multiply a fraction by a whole number means to multiply numerator by the whole number and keep the denominator same.

3.2.2 Multiply a fraction by another fraction

Consider the following examples:

Example 1. Multiply $\frac{1}{2}$ by $\frac{4}{5}$.

Solution:

We know that, $\frac{1}{2}$ by $\frac{4}{5}$ means $\frac{1}{2} \times \frac{4}{5}$

$$\begin{aligned} \text{So, } \frac{1}{2} \times \frac{4}{5} \\ = \frac{1 \times 4}{2 \times 5} = \frac{2}{5} \end{aligned}$$

2
1

To multiply one fraction by another fraction, we multiply the numerator with numerator and the denominator with denominator. We can simplify a product by cancelling the common factors. Here 2 is a common factor of denominator 2 and numerator 4 so it is cancelled out.

Example 2. Multiply $\frac{1}{7}$ by $\frac{5}{6}$.

Solution: $\frac{1}{7}$ by $\frac{5}{6}$ means $\frac{1}{7} \times \frac{5}{6}$

$$\frac{1}{7} \times \frac{5}{6} = \frac{1 \times 5}{7 \times 6} = \frac{5}{42}$$

Example 3. Simplify.

(a) $2 \frac{2}{3} \times \frac{9}{4}$

Solution:

$$2 \frac{2}{3} \times \frac{9}{4} = \frac{8}{3} \times \frac{9}{4}$$

$$= \frac{\overset{2}{\cancel{8}} \times \overset{3}{\cancel{9}}}{\underset{1}{\cancel{3}} \times \underset{1}{\cancel{4}}}$$

$$= \frac{2 \times 3}{1 \times 1}$$

$$= \frac{6}{1} = 6$$

(b) $\frac{6}{5} \times \frac{25}{8} \times \frac{1}{3}$

Solution:

$$\frac{6}{5} \times \frac{25}{8} \times \frac{1}{3}$$

$$= \frac{\overset{1}{\cancel{6}} \times \overset{5}{\cancel{25}} \times 1}{\underset{1}{\cancel{5}} \times \underset{4}{\cancel{8}} \times \underset{1}{\cancel{3}}}$$

$$= \frac{1 \times 5 \times 1}{1 \times 4 \times 1}$$

$$= \frac{5}{4} = 1 \frac{1}{4}$$

EXERCISE 3.2

A. Multiply the fraction by the given whole number. Also demonstrate with the help of diagram.

(1) $\frac{3}{4} \times 4$

(2) $\frac{1}{3} \times 3$

(3) $\frac{3}{5} \times 5$

(4) $\frac{1}{2} \times 2$

(5) $\frac{1}{4} \times 8$

(6) $\frac{1}{3} \times 6$

(7) $\frac{3}{2} \times 4$

(8) $\frac{2}{3} \times 6$

(9) $\frac{3}{5} \times 11$

B. Multiply the given fraction by another fraction.

(1) $\frac{1}{2}$ by $\frac{1}{5}$ (2) $\frac{1}{4}$ by $\frac{3}{5}$ (3) $\frac{1}{3}$ by $\frac{2}{3}$

(4) $1\frac{1}{4}$ by $\frac{3}{4}$ (5) $2\frac{1}{3}$ by $\frac{4}{5}$ (6) $3\frac{1}{2}$ by $1\frac{1}{2}$

C. Simplify.

(1) $\frac{3}{4} \times \frac{8}{9}$ (2) $\frac{2}{5} \times \frac{25}{4}$ (3) $\frac{7}{3} \times \frac{9}{14}$

(4) $1\frac{5}{18} \times 2\frac{1}{10} \times \frac{8}{7}$ (5) $1\frac{6}{7} \times \frac{14}{15} \times 3\frac{1}{8}$ (6) $2\frac{1}{3} \times 1\frac{2}{3} \times \frac{20}{7}$

Multiply two or more fractions involving brackets (proper, improper and mixed fractions)

Let us see the following examples:

Example 1. Multiply $\frac{6}{8}$ by $\frac{19}{6}$.

Solution: $\frac{6}{8} \times \frac{19}{6} = \frac{\cancel{6}^1 \times 19}{8 \times \cancel{6}_1} = \frac{19}{8} = 2\frac{3}{8}$

Example 2. Solve: $1\frac{1}{4} \times \left(\frac{5}{10} \times \frac{4}{5}\right)$

Solution: $1\frac{1}{4} \times \left(\frac{5}{10} \times \frac{4}{5}\right) = \frac{5}{4} \times \left(\frac{5}{10} \times \frac{4}{5}\right)$

Remember to solve the brackets first!

$$\begin{aligned} &= \frac{5}{4} \times \left(\frac{\cancel{5}^1 \times \cancel{4}^2}{\cancel{10}_5 \times \cancel{5}_1}\right) \\ &= \frac{5}{4} \times \left(\frac{1 \times 2}{5 \times 1}\right) = \frac{5}{4} \times \frac{2}{5} \\ &= \frac{\cancel{5}^1 \times \cancel{2}^1}{\cancel{4}_2 \times \cancel{5}_1} = \frac{1}{2} \end{aligned}$$

Thus, $1\frac{1}{4} \times \left(\frac{5}{10} \times \frac{4}{5}\right) = \frac{1}{2}$

Example 3. Solve: $1\frac{1}{2} \times \left(2\frac{1}{4} \times 3\frac{1}{3}\right)$

Solution: $1\frac{1}{2} \times \left(2\frac{1}{4} \times 3\frac{1}{3}\right)$

First convert mixed fractions into improper fractions

$$\begin{aligned}
 1\frac{1}{2} \times \left(2\frac{1}{4} \times 3\frac{1}{3}\right) &= \frac{3}{2} \times \frac{9}{4} \times \frac{10}{3} \\
 &= \frac{3}{2} \times \left(\frac{\overset{3}{\cancel{9}} \times \overset{5}{\cancel{10}}}{\underset{2}{\cancel{4}} \times \underset{1}{\cancel{3}}}\right) \\
 &= \frac{3}{2} \times \left(\frac{3 \times 5}{2 \times 1}\right) \\
 &= \frac{3}{2} \times \frac{15}{2} \\
 &= \frac{3 \times 15}{2 \times 2} = \frac{45}{4} = 11\frac{1}{4}
 \end{aligned}$$

Thus, $1\frac{1}{2} \times \left(2\frac{1}{4} \times 3\frac{1}{3}\right) = 11\frac{1}{4}$

Example 4. Simplify $\left(\frac{5}{9} \times \frac{6}{11}\right) \times \frac{21}{10}$

$$\begin{aligned}
 \text{Solution: } \left(\frac{5}{9} \times \frac{6}{11}\right) \times \frac{21}{10} &= \left(\frac{\overset{2}{\cancel{6}} \times 5}{\underset{3}{\cancel{9}} \times 11}\right) \times \frac{21}{10} \\
 &= \left(\frac{5 \times 2}{3 \times 11}\right) \times \frac{21}{10} = \frac{10}{33} \times \frac{21}{10} \\
 &= \frac{\overset{1}{\cancel{10}} \times \overset{7}{\cancel{21}}}{\underset{11}{\cancel{33}} \times \underset{1}{\cancel{10}}} = \frac{1 \times 7}{11 \times 1} \\
 &= \frac{7}{11}
 \end{aligned}$$

Thus, $\left(\frac{5}{9} \times \frac{6}{11}\right) \times \frac{21}{10} = \frac{7}{11}$

EXERCISE 3.3

Solve.

(1) $\frac{5}{6} \times 2\frac{1}{4} \times \frac{16}{5}$

(2) $\frac{7}{9} \times \left(2\frac{1}{4} \times \frac{8}{7}\right)$

(3) $\left(\frac{4}{5} \times \frac{10}{3}\right) \times \frac{9}{8}$

(4) $\left(\frac{3}{4} \times \frac{16}{5}\right) \times 1\frac{2}{3}$

(5) $\left(1\frac{3}{5} \times 10\frac{1}{2}\right) \times \frac{5}{21}$

(6) $\left(1\frac{1}{6} \times \frac{5}{6}\right) \times 5\frac{1}{7}$

(7) $1\frac{5}{16} \times \left(12\frac{1}{2} \times 1\frac{11}{21}\right)$

(8) $2\frac{5}{6} \times \left(1\frac{3}{17} \times 2\frac{1}{10}\right)$

(9) $\left(3\frac{9}{10} \times \frac{20}{36}\right) \times 1\frac{11}{13}$

(10) $\left(2\frac{4}{5} \times 1\frac{5}{7}\right) \times 2\frac{1}{12}$

(11) $4\frac{2}{7} \times \left(2\frac{5}{8} \times 1\frac{1}{9}\right)$

(12) $8\frac{1}{3} \times \left(2\frac{1}{10} \times \frac{1}{7}\right)$

3.2.4 Verify Distributive Laws:

There are two laws of distributive multiplication.

- (1) Distributive law of multiplication over addition
- (2) Distributive law of multiplication over subtraction

1. Distributive law of multiplication over addition

If $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{1}{4}$ are any three fractions, thenConsider $\frac{1}{2} \times \left(\frac{2}{3} + \frac{1}{4}\right)$ and $\frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{1}{4}$

Let us verify it:

$$\frac{1}{2} \times \left(\frac{2}{3} + \frac{1}{4}\right) = \left(\frac{1}{2} \times \frac{2}{3}\right) + \left(\frac{1}{2} \times \frac{1}{4}\right)$$

$$\frac{1}{2} \times \left(\frac{8+3}{12}\right) = \frac{2}{6} + \frac{1}{8}$$

$$\frac{1}{2} \times \frac{11}{12} = \frac{8+3}{24}$$

$$\frac{11}{24} = \frac{11}{24}$$

$$\text{So, LHS} = \text{RHS}$$

$$\text{Therefore, } \frac{1}{2} \times \left(\frac{2}{3} + \frac{1}{4} \right) = \left(\frac{1}{2} \times \frac{2}{3} \right) + \left(\frac{1}{2} \times \frac{1}{4} \right)$$

This indicates distributive law of multiplication over addition.

$$\text{Now verify it: } \left(\frac{2}{3} + \frac{1}{4} \right) \times \frac{1}{2} = \left(\frac{2}{3} \times \frac{1}{2} \right) + \left(\frac{1}{4} \times \frac{1}{2} \right)$$

2. Distributive law of multiplication over subtraction

If $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ are any three fractions, then the distributive law says that the two expressions below will be equal.

$$\frac{1}{2} \times \left(\frac{1}{3} - \frac{1}{4} \right) \text{ and } \left(\frac{1}{2} \times \frac{1}{3} \right) - \left(\frac{1}{2} \times \frac{1}{4} \right)$$

$$\text{Let us verify it: } \frac{1}{2} \times \left(\frac{1}{3} - \frac{1}{4} \right) = \left(\frac{1}{2} \times \frac{1}{3} \right) - \left(\frac{1}{2} \times \frac{1}{4} \right)$$

$$\frac{1}{2} \times \left(\frac{4-3}{12} \right) = \frac{1}{6} - \frac{1}{8}$$

$$\frac{1}{2} \times \frac{1}{12} = \frac{4-3}{24}$$

$$\frac{1}{24} = \frac{1}{24}$$

$$\text{So, LHS} = \text{RHS}$$

$$\text{Therefore, } \frac{1}{2} \times \left(\frac{1}{3} - \frac{1}{4} \right) = \left(\frac{1}{2} \times \frac{1}{3} \right) - \left(\frac{1}{2} \times \frac{1}{4} \right)$$

This indicates distributive law of multiplication over subtraction.

$$\text{Now verify it: } \left(\frac{1}{3} - \frac{1}{4} \right) \times \frac{1}{2} = \left(\frac{1}{3} \times \frac{1}{2} \right) - \left(\frac{1}{4} \times \frac{1}{2} \right)$$

EXERCISE 3.4

A. Verify the distributive law of multiplication over addition in the following.

$$(1) \quad \frac{2}{5} \times \left(\frac{3}{7} + \frac{4}{5} \right) = \left(\frac{2}{5} \times \frac{3}{7} \right) + \left(\frac{2}{5} \times \frac{4}{5} \right)$$

$$(2) \quad \frac{7}{9} \times \left(\frac{1}{4} + \frac{1}{3} \right) = \left(\frac{7}{9} \times \frac{1}{4} \right) + \left(\frac{7}{9} \times \frac{1}{3} \right)$$

$$(3) \quad \frac{2}{7} \times \left(\frac{3}{8} + \frac{1}{5} \right) = \left(\frac{2}{7} \times \frac{3}{8} \right) + \left(\frac{2}{7} \times \frac{1}{5} \right)$$

$$(4) \quad \left(\frac{1}{9} + \frac{4}{9} \right) \times \frac{3}{4} = \left(\frac{1}{9} \times \frac{3}{4} \right) + \left(\frac{4}{9} \times \frac{3}{4} \right)$$

B. Verify the distributive law of multiplication over subtraction in the following.

$$(1) \quad \frac{1}{2} \times \left(\frac{3}{4} - \frac{2}{3} \right) = \left(\frac{1}{2} \times \frac{3}{4} \right) - \left(\frac{1}{2} \times \frac{2}{3} \right)$$

$$(2) \quad \frac{1}{5} \times \left(\frac{1}{2} - \frac{1}{3} \right) = \left(\frac{1}{5} \times \frac{1}{2} \right) - \left(\frac{1}{5} \times \frac{1}{3} \right)$$

$$(3) \quad \left(\frac{4}{5} - \frac{3}{4} \right) \times \frac{2}{3} = \left(\frac{4}{5} \times \frac{2}{3} \right) - \left(\frac{3}{4} \times \frac{2}{3} \right)$$

$$(4) \quad \frac{5}{6} \times \left(\frac{4}{7} - \frac{1}{2} \right) = \left(\frac{5}{6} \times \frac{4}{7} \right) - \left(\frac{5}{6} \times \frac{1}{2} \right)$$

Solve real life problems involving multiplication of fractions

Example 1. There were 56 students in class five. On a rainy day $\frac{1}{8}$ of them were absent. How many were present?

Solution: Total number of students = 56

Number of students absent on rainy day = $\frac{1}{8}$ of 56 = $\frac{1}{8} \times 56$

$$= \frac{1}{8} \times \frac{56}{1} = \frac{1 \times 56}{8 \times 1} = \frac{1 \times 7}{1 \times 1} = 7$$

Total number of students	—	Number of students absent	=	Number of students present
56	—	7	=	49

Hence 49 students were present on rainy day.

Example 2.

The engineers drilled $\frac{5}{6}$ km of a tunnel in January and drilled only $\frac{1}{6}$ of the previously drilled in February. What fraction of the tunnel did they drill in February?

Solution:

Engineer drilled in January = $\frac{5}{6}$ km of a tunnel

They drilled in February $\frac{1}{6}$ of the $\frac{5}{6}$ km in January

$$= \frac{1}{6} \times \frac{5}{6} \text{ km} = \frac{5}{36} \text{ km}$$

Thus they drilled $\frac{5}{36}$ km in February

EXERCISE 3.5

1. The height of a door is $2\frac{2}{3}$ metres. Out of which $\frac{1}{8}$ the part was trimmed off. How much part was trimmed off?
2. On Monday one-tenth of the total number of students were on leave. If the total number of the students are enrolled 550. How many students were present?
3. A surgeon advised an operating room technician to arrange 12 drips of dextrose solution. The hospital has only $\frac{2}{3}$ of the required dextrose. How many drips are available in the hospital?

4. A lady used $\frac{1}{5}$ of 50 kg of wheat for making bread. If she makes bread twice a day, then how much wheat will be used?
5. Find the price of 13 dozen of eggs; if the price of one egg is Rs $9\frac{1}{2}$
6. A lady bought $5\frac{2}{5}$ metres of cloth. She used $\frac{1}{3}$ of the cloth in making a TV cover. How much cloth she used?
7. A cooler holds 8 bottles of water. If one bottle is holding $2\frac{1}{2}$ litre water, find the capacity of full cooler.
8. What is the total length of the cloth? If it consists of 12 pieces of cloth and each piece is of length $1\frac{1}{2}$ metres.
9. The cost of 1 kg potatoes is Rs 45. Find the cost of $15\frac{2}{4}$ kg of that potatoes.
10. A piece of wire is $8\frac{1}{3}$ metre long. Find the total length of wire, if there are $13\frac{1}{5}$ such small pieces.

3.3 DIVISION

3.3.1 Divide a fraction by a number

If we divide a circle into four equal parts.
How many quarters are in one whole?

That is $\frac{1}{1} \div \frac{1}{4} = \boxed{4}$



Or $\frac{1}{1} \times \frac{4}{1} = \frac{1 \times 4}{1 \times 1} = 4$ quarters.

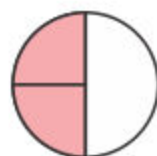
Here, $\frac{4}{1}$ is reciprocal of $\frac{1}{4}$

Again how many quarters in one half?

Teacher's Note

Teacher should explain the students to divide a fraction by a whole number we need to multiply by the reciprocal of the numbers.

That is $\frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \times \frac{4}{1}$
 $= \frac{1 \times 4}{2 \times 1} = 2 \text{ quarters}$



Let us consider the following examples:

Example 1. Simplify: $\frac{1}{5} \div 3$

Solution: $\frac{1}{5} \div 3 = \frac{1}{5} \div \frac{3}{1} \quad \left(\frac{1}{5} \div \frac{3}{1} \text{ means } \frac{1}{5} \times \frac{1}{3}\right)$
 $= \frac{1}{5} \times \frac{1}{3} \quad \left(\frac{1}{3} \text{ is reciprocal of } 3\right)$
 $= \frac{1 \times 1}{5 \times 3} = \frac{1}{15}$

Similarly, simplifying $\frac{1}{3} \div 5$

$$\frac{1}{3} \div 5 = \frac{1}{3} \times \frac{1}{5} \quad \left(\frac{1}{5} \text{ is reciprocal of } 5\right)$$

$$= \frac{1 \times 1}{5 \times 3} = \frac{1}{15}$$

Example 2. Divide $1\frac{5}{9}$ **by** 7

Solution:

First we change the mixed fraction into an improper fraction.

$$\frac{14}{9} \div 7$$

$$= \frac{14}{9} \times \frac{1}{7} \quad \left(\frac{1}{7} \text{ is reciprocal of } 7\right)$$

$$= \frac{14 \times 1}{9 \times 7} = \frac{2 \times 7 \times 1}{9 \times 7} = \frac{2}{9}$$

Thus, $1\frac{5}{9} \div 7 = \frac{2}{9}$

To divide a fraction by a non-zero whole number, we multiply the fraction by the reciprocal of the whole number.

EXERCISE 3.6

A. Write the reciprocals of each of the following.

(1) 2 (2) 4 (3) $\frac{1}{3}$ (4) $\frac{1}{5}$ (5) $\frac{1}{10}$

(6) $\frac{3}{4}$ (7) $2\frac{6}{7}$ (8) $1\frac{8}{5}$ (9) $2\frac{1}{3}$ (10) $4\frac{2}{3}$

B. Solve.

(1) $\frac{1}{2} \div \frac{5}{6}$ (2) $\frac{2}{3} \div 4$ (3) $\frac{3}{4} \div 6$ (4) $\frac{4}{5} \div 8$

(5) $\frac{5}{6} \div 10$ (6) $\frac{4}{5} \div 12$ (7) $\frac{7}{8} \div 14$ (8) $\frac{8}{9} \div 24$

(9) $2\frac{2}{5} \div 36$ (10) $1\frac{1}{9} \div 20$ (11) $2\frac{1}{5} \div 55$ (12) $3\frac{1}{3} \div 5$

3.3.2 Divide a fraction by another fraction (proper, improper and mixed)

To divide a fraction by another fraction is to multiply the first fraction by the reciprocal of another given fraction. Consider the following examples.

Example 1. Divide $\frac{9}{16}$ by $\frac{3}{4}$

Solution: $\frac{9}{16} \div \frac{3}{4} = \frac{9}{16} \times \frac{4}{3}$ ($\frac{4}{3}$ is reciprocal of $\frac{3}{4}$)

$$= \frac{9 \times 4}{16 \times 3} = \frac{\overset{3}{\cancel{9}} \times \overset{4}{\cancel{4}}}{\underset{4}{\cancel{16}} \times \underset{3}{\cancel{3}}}$$

$$= \frac{3 \times 1}{4 \times 1} = \frac{3}{4}$$

Example 2. Divide $\frac{7}{4}$ by $\frac{5}{4}$

Solution: $\frac{7}{4} \div \frac{5}{4} = \frac{7}{4} \times \frac{4}{5}$
 $= \frac{7 \times 4}{4 \times 5} = \frac{7 \times \cancel{4}^1}{\cancel{4}_1 \times 5} = \frac{7}{5} = 1\frac{2}{5}$

Hence, $\frac{7}{4} \div \frac{5}{4} = 1\frac{2}{5}$

Example 3. Divide $2\frac{5}{8}$ by $2\frac{1}{3}$

Solution: First we change mixed fraction into improper fraction.

$$\begin{aligned} 2\frac{5}{8} \div 2\frac{1}{3} &= \frac{2 \times 8 + 5}{8} \div \frac{2 \times 3 + 1}{3} \\ &= \frac{16 + 5}{8} \div \frac{6 + 1}{3} = \frac{21}{8} \div \frac{7}{3} = \frac{21}{8} \times \frac{3}{7} \\ &= \frac{\overset{3}{\cancel{21}} \times 3}{8 \times \cancel{7}_1} = \frac{3 \times 3}{8 \times 1} = \frac{9}{8} = 1\frac{1}{8} \end{aligned}$$

Thus, $2\frac{5}{8} \div 2\frac{1}{3} = 1\frac{1}{8}$

EXERCISE 3.7

Solve the following.

(1) $\frac{1}{2} \div \frac{3}{4}$

(2) $\frac{7}{6} \div \frac{4}{9}$

(3) $\frac{3}{4} \div \frac{15}{16}$

(4) $\frac{9}{7} \div \frac{81}{14}$

(5) $\frac{1}{6} \div \frac{1}{12}$

(6) $\frac{11}{22} \div \frac{11}{22}$

(7) $\frac{16}{9} \div \frac{4}{3}$

(8) $\frac{20}{30} \div \frac{40}{30}$

(9) $1\frac{1}{4} \div \frac{1}{4}$

(10) $\frac{2}{3} \div 4\frac{3}{4}$

(11) $8\frac{1}{2} \div 3\frac{1}{2}$

(12) $2\frac{1}{5} \div 1\frac{1}{6}$

(13) $9\frac{3}{5} \div 2\frac{1}{4}$

(14) $3\frac{8}{9} \div 1\frac{1}{9}$

(15) $10\frac{1}{9} \div 4\frac{3}{4}$

Solve real life problems involving division of fractions

Example 1. How many half metre pieces can be cut from a stick which is $12\frac{1}{2}$ metres long?

Solution: Length of stick = $12\frac{1}{2}$ m (Dividend $12\frac{1}{2}$ a mixed fraction)

Length of one of the required piece = $\frac{1}{2}$ m (Divisor $\frac{1}{2}$ is proper fraction)

$$\begin{aligned}\text{Total numbers of pieces} &= 12\frac{1}{2} \div \frac{1}{2} = \frac{25}{2} \div \frac{1}{2} \\ &\quad (\text{Reciprocal of } \frac{1}{2} \text{ is } \frac{2}{1}) \\ &= \frac{25}{2} \times \frac{2}{1} = \frac{25 \times \overset{1}{\cancel{2}}}{\underset{1}{\cancel{2}} \times 1} = \boxed{25}\end{aligned}$$

Thus, we get **25** pieces.

EXERCISE 3.8

1. Talib buys $10\frac{1}{2}$ kg tomatoes for Rs 210. What is the price of 1 kg of tomatoes?
2. One-fifth of the 35 cars in the parking area are blue. How many blue cars are there in the parking area?
3. Naila buys $\frac{7}{3}$ m piece of a lace. She wants to make its small pieces at length $\frac{1}{12}$ m. How many pieces can be made?
4. The camp cook made $1\frac{3}{4}$ kg of baked beans. Each serving of beans is $\frac{1}{4}$ kg. How many servings of beans did the cook make?

5. A road is $50\frac{1}{2}$ metre long. Half of it was damaged in rain. How much road is safe?
6. Rehan bought $82\frac{1}{2}$ metre cloth. He used this cloth for making uniform of 15 children of equal size. How much is cloth used in each dress?
7. A plastic drum holds $49\frac{1}{2}$ litres drinking water. How many bottles holding $1\frac{1}{2}$ litres of drinking water can be filled from it?
8. A boy completes $8\frac{3}{4}$ km in $2\frac{1}{2}$ hours on bicycle in a race. How much distance does he complete in 1 hour?

3.4 SIMPLIFY EXPRESSIONS INVOLVING FRACTIONS USING BODMAS RULE

We know that in BODMAS, B stands for Brackets, O stands for Of, D stands for Division, M stands for Multiplication, A stands for Addition and S stands for Subtraction.

Example 1: Simplify by using BODMAS rule:

$$\frac{5}{3} \times \left(1\frac{1}{3} - \frac{1}{2}\right) \div \frac{5}{2}$$

Solution:

Solve operations within 'brackets'.

$$\frac{5}{3} \times \left(1\frac{1}{3} - \frac{1}{2}\right) \div \frac{5}{2} = \frac{5}{3} \times \left(\frac{4}{3} - \frac{1}{2}\right) \div \frac{5}{2}$$

By following the order of BODMAS, we first simplify brackets then of, and then arithmetic operations. In arithmetic operations first Division next Multiplication, and in last Addition and Subtraction.

$$= \frac{5}{3} \times \left(\frac{8-3}{6}\right) \div \frac{5}{2} \quad (\text{By taking LCM})$$

$$= \frac{5}{3} \times \frac{5}{6} \div \frac{5}{2}$$

$$= \frac{5}{3} \times \frac{5}{6} \times \frac{2}{5} \left(\frac{2}{5} \text{ is reciprocal of } \frac{5}{2} \right)$$

$$= \frac{5 \times \cancel{5}^1 \times \cancel{2}_1}{3 \times \cancel{6}_3 \times \cancel{5}_1} = \frac{5 \times 1 \times 1}{3 \times 3 \times 1} = \frac{5}{9}$$

Example 2. Simplify $\frac{1}{2}$ of $\frac{4}{5} + \left(2\frac{1}{3} - 1\frac{1}{4} \right)$

Solution: First solve within bracket.

$$\begin{aligned} \frac{1}{2} \text{ of } \frac{4}{5} + \left(2\frac{1}{3} - 1\frac{1}{4} \right) &= \frac{1}{2} \text{ of } \frac{4}{5} + \left(2\frac{1}{3} - 1\frac{1}{4} \right) \\ &= \frac{1}{2} \times \frac{4}{5} + \left(\frac{3 \times 2 + 1}{3} - \frac{4 \times 1 + 1}{4} \right) \end{aligned}$$

By simplifying

$$\begin{aligned} &= \frac{1 \times \cancel{4}^2}{\cancel{2}_1 \times 5} + \left(\frac{7}{3} - \frac{5}{4} \right) \\ &= \frac{1 \times 2}{1 \times 5} + \left(\frac{28 - 15}{12} \right) \\ &= \frac{2}{5} + \frac{13}{12} \quad \text{By taking LCM of 5 and 12} \\ &= \frac{24 + 65}{60} \\ &= \frac{89}{60} \\ &= 1\frac{29}{60} \end{aligned}$$

EXERCISE 3.9

Simplify the followings by using BODMAS rule.

(1) $\frac{3}{4} + \frac{2}{9} \times 4\frac{1}{3} \div 3\frac{1}{4}$

(2) $\frac{1}{4} \times \left(\frac{8}{3} + \frac{2}{7}\right)$

(3) $\frac{1}{2} + \left(\frac{3}{4} \times 1\frac{7}{33} - \frac{1}{3}\right)$

(4) $\left(3\frac{1}{6} - 1\frac{1}{4}\right) \times 2$

(5) $\left(\frac{4}{5} - \frac{3}{10}\right) \times \left(\frac{1}{2} + \frac{3}{4}\right)$

(6) $\left(2\frac{1}{2} \div \frac{3}{4}\right) \times \frac{3}{7} - \left(\frac{1}{4} + \frac{1}{8}\right)$

(7) $1\frac{3}{5} \times \left(\frac{4}{3} - \frac{3}{4} + 2\frac{1}{3}\right) \div 1\frac{2}{7}$

(8) $1\frac{1}{6} + \left(2\frac{3}{4} \times 3\frac{1}{3} \div 2\frac{1}{4}\right) - 7\frac{1}{2}$

(9) $\left(1\frac{3}{5} \text{ of } \frac{5}{6}\right) - \left(2\frac{3}{7} + 1\frac{1}{5}\right)$

(10) $\left(\frac{5}{4} + \frac{8}{3}\right) \times \left(\frac{10}{3} + \frac{5}{2}\right)$

(11) $20 + 5 \text{ of } 9 - \left(1\frac{2}{3} \times \frac{1}{5}\right)$

(12) $4\frac{1}{2} + \left(5\frac{1}{3} \text{ of } 3\right) - 2\frac{2}{3}$

REVIEW EXERCISE 3

(1) Write 'T' for true and 'F' for false.

(i) $4\frac{2}{3}$ is an example of an improper fraction.

☐

(ii) $\frac{1}{3} \div \frac{3}{1} = \frac{1}{9}$

☐

(iii) The lowest form of $\frac{12}{4} = \frac{1}{3}$

☐

(iv) $\frac{3}{4} - \frac{1}{2} = \frac{1}{4}$

(v) $\frac{1}{3}, \frac{2}{6}, \frac{3}{9}$ are equivalent fractions.

(vi) The reciprocal of $2\frac{1}{2}$ is $\frac{2}{5}$.

(vii) $\frac{3}{4}$ of a dozen is 9.

(viii) $1\frac{1}{2}$ m equal to 150 cm.

(2) Solve the following.

(i) Five ball points are bought for Rs 50 $\frac{3}{4}$. Find the price of one ball point.

(ii) Afzal and Fazila bought a chocolate bar. Afzal got $\frac{2}{5}$ and Fazila got $\frac{3}{10}$. Who got more and by how much?

(iii) Find the price of 2 dozen packets of biscuit, if the price of one packet is Rs $12\frac{1}{2}$.

(iv) Shumaila buys $2\frac{1}{4}$ m ribbon. She used $1\frac{7}{8}$ m of it. How much ribbon is left?

(v) $75\frac{3}{4}$ kg of tea is packed into packets. Each packet contains $\frac{3}{8}$ kg of tea. How many packets are needed to fill the tea?

(vi) At a charity show each student in a school purchases a ticket costing Rs 30. If $\frac{3}{4}$ the amount collected is Rs 24600. Find the number of tickets sold.

(vii) Simplify: (a) $2\frac{1}{3} - \left(\frac{16}{5} \div 1\frac{7}{8} \text{ of } 2\frac{2}{15}\right) + 1\frac{1}{8}$

(b) $\frac{3}{4} + \frac{3}{4} \text{ of } \frac{3}{4} \div \frac{3}{4} - \left(\frac{3}{4} \times \frac{3}{4}\right)$